Why I Want to Join HackerFab IITB

Nakul Ramji | B.Tech, Metallurgical Engineering and Materials Science, IIT Bombay | nakulramji@iitb.ac.in

I want to join HackerFab IITB because I learn best by building, and this is one of the few places where a student can take a wafer through every step, from oxidation and lithography to deposition, and then measure what was actually made. After reading the documentation, what stood out to me is how practical the team's approach is: tools are built from salvaged and locally sourced parts on tight budgets, the documentation is honest about what is still marked TODO, and the wider goal of showing that low-cost, student-run fabrication works in India is one I want to contribute to.

I'm a second-year Metallurgical Engineering and Materials Science student with no fab or research experience yet, but with relevant hands-on experience. I built a wireless dump-truck prototype with two teammates, owning the joystick controller's CAD, 3D printing, soldering and integration. In my Sensors and Measurements Lab project, I'm part of a team building a calibrated TCS230 colourimeter with a light-tight 45°/0° geometry and dark/white normalisation, which we are using to estimate the tempering temperature of steel from oxide heat-tint colour. I have also built two AI-assisted software projects, which taught me to design, specify and debug data pipelines. I'm comfortable with Arduino, sensor interfacing, calibration and CAD.

Short term. In my first few months, I want to learn the basics properly: wafer handling, cleaning, and running existing recipes safely on the litho system and the tube furnace. I also want to take ownership of something small and finish it. For that I am exploring a repeatable, low-cost hot plate, since the litho recipes depend on bakes at 90°C and 150°C, and bake temperature and uniformity directly affect resist performance. Alongside, I'd like to help with litho-v1's open measurements (isolated line widths, stage repeatability, overlay error) and the LED current-to-intensity mapping, so that dose matrices become repeatable. For every tool I touch, I'll write the missing SOPs and run logs.

Long term. I want to help the team reach a working NMOS transistor by following the roadmap in the process documentation: reliable MOS capacitors to prove oxide quality, large diodes to prove doping and contacts, clean liftoff patterns, then micron-scale devices and finally an aligned three-mask NMOS. I'm especially interested in the materials side of this: oxide quality, getting photoresist to survive HF (adhesion promotion, a chromium hard mask, buffered oxide etch), contact metallurgy, and thin-film characterisation with ellipsometry and profilometry. Eventually, I'd like to own a tool end to end, for example by upgrading the sputter coater with a full SOP, interlocks, automation and a thickness monitor.

Where I'd like to take the team. I'd like us to keep a run-by-run record in which every run is logged with its parameters, images and measurements, so that recipes improve from data rather than memory. I'd push for complete, tested SOPs and checklists for every tool, for sharing our designs and results back with the wider HackerFab community, and for helping other colleges replicate our tools so that student-run fabs spread across India.

What I hope to gain. I want hands-on fabrication skill, the habit of characterising and debugging from first principles, a solid grounding in operating hazardous equipment safely, and guidance from people who have already solved the problems I'm about to meet. In return, I'll commit to regular tool runs, careful records, and the unglamorous work of documentation and helping newer members get started.